

Md. Abu Bokkor Shiddik

Researcher | Bachelor of Science in Statistics

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Link: [Portfolio](#) | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#) | [GitHub](#) | [Web of Science](#) | [Scopus ID](#)

RESEARCH INTEREST

I have a strong interest in using data science, machine learning and explainable AI to improve global health, especially in disease modeling and climate–health decision making.

SKILLS

Machine Learning, Deep Learning & AI

Regression, classification, decision trees, random forest, gradient boosting (XGBoost, LightGBM, CatBoost), SVM, KNN, Naive Bayes, clustering, PCA, neural networks (ANN, CNN, RNN, LSTM, GRU)

Explainable AI & Model Interpretation

SHAP, LIME, Integrated Gradients, Layer-wise Relevance Propagation, feature importance analysis

Statistical, Spatio-Temporal & Health Analytics

Hypothesis testing, ANOVA, GLM, ARIMA/SARIMA, Bayesian inference, spatio-temporal modeling, hotspot analysis (Gi*), GIS (sf, raster), epidemiological & disease prediction modeling, climate–health analysis

Programming & Data Tools

R, Python, C, Java, SPSS, LaTeX, Git/GitHub, HTML

Research Practice

Scientific writing, systematic review, reproducible research

EDUCATION

March, 2022 -June, 2026	Bachelor of Science in Statistics Begum Rokeya University, Rangpur, Bangladesh Session: 2020-21; Result: CGPA: 3.72 out of 4.00
January, 2018 - December, 2019	Higher Secondary Certificate (Science)-2019 Police Lines School and College, Rangpur, Bangladesh Session: 2017-18; Result: GPA: 5.00 out of 5.00
January, 2015 - December, 2016	Dakhil (SSC) Certificate (Science)-2017 Pakuria Sharif Se. Fagil Madrasah, Rangpur, Bangladesh Result: GPA: 5.00 out of 5.00

CONFERENCE & WORKSHOP

International Conference on Applied Statistics and Data Science-2025, University of Dhaka

Abstract ID: P-145 & P-105 [↗](#)

Presented research work: Spatio-temporal modeling of global tuberculosis prediction based on unified-machine learning framework [↗](#)

Workshop on ‘Promotion of Research in Higher Education (10 march, 2025)

Institutional Quality Assurance Cell (IQAC), Begum Rokeya University, Rangpur.

CERTIFICATIONS

March, 2026	Certificate of Publication The certificate of publication for the article titled: District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning; <i>Trop. Med. Infect. Dis.</i> 2026, Volume 11, Issue 3, 73
March, 2026	Certificate of Excellence; Research design by Researcher Academy, ELSEVIER ↗
23 March 2026	Certificate of Appreciation in recognition of peer reviewer contributions for BMJ Global Health by BMJ ↗

16 February 2026	Certificate of Appreciation in recognition of peer reviewer contributions for BMJ Public Health by BMJ ↗
January, 2026	Certificate of Excellence; Technical writing skills by Researcher Academy, ELSEVIER ↗
December, 2025	Certificate of Appreciation for presenting a research paper at the Poster session of the International Conference on Applied Statistics and Data Science (ICASDS) by Institute of Statistical Research and Training (ISRT), University of Dhaka ↗

COURSES AND TRAINING

January - March, 2026	Scientific & Manuscript Writing Series (6 modules: Gen AI use in the research workflow ↗ , interdisciplinary writing ↗ , case reports ↗ , identifying research gaps ↗ , LaTeX manuscripts ↗ , methods articles ↗ , review articles ↗) by Researcher Academy, Elsevier
15 March, 2025	Practical Time Series Analysis An online course by The State University of New York ↗
10 September, 2024	Introduction to Climate Risk Informed Decision Analysis (CRIDA) a course of study offered by UNESCO ↗
January – Jun 2024	Data Science & Machine Learning Specialization (6 courses: Machine Learning with Python ↗ ↗ , Data Visualization with R ↗ , Regression for Car Mileage/Diamond Price ↗ , R for Data Science ↗ , Data Science Methodology ↗ , What is Data Science ↗) by IBM
10 March, 2024	Successful Presentation An online course by University of Colorado Boulder ↗
December 2023	Research Methodology Series (4 courses: Interdisciplinary Research Methodology ↗ , Research Methodologies ↗ , Understanding Research Methods ↗ , Introduction to Computer Programming ↗) by RSBRU / Queen Mary University of London / University of London
24 December, 2023	Introduction to Statistical Analysis: Hypothesis Testing By SAS ↗
November, 2023	UNICEF Training Series (2 courses: Education in Emergencies ↗ , Leadership for Planning and Decision-Making with Adolescents ↗)
July, 2016-December, 2016	Deploma in Computer Science A six month training course by Bangladesh Computer Education (BCE) ↗

RESEARCH PROJECT

Project Title: Exploring community knowledge, health belief and healthcare-seeking behaviours regarding Non-Communicable Diseases (NCD's) in Bangladesh[↗](#)
Funding Source: University Grant Commission; **Role:** Research Assistant (Start date: Oct.2025 & end date: Feb.2026) , Department of Statistics, Begum Rokeya University, Rangpur

Project Title: District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning [↗](#)
Description: A project submitted in partial fulfillment of the requirements for the degree of B.Sc. In Statistics, Begum Rokeya University, Rangpur.

RESEARCH & PUBLICATION [↗](#)

1. Rahman, M.S., **Shiddik, A.B.** Utilizing artificial intelligence to predict and analyze socioeconomic, environmental, and healthcare factors driving tuberculosis globally. *Sci Rep* 15, 13619 (2025). <https://doi.org/10.1038/s41598-025-96973-w>
2. Rahman, M. S., Amrin, M., & **Bokkor Shiddik, M. A.** (2025). Dengue Early Warning System and Outbreak Prediction Tool in Bangladesh Using Interpretable Tree-Based Machine Learning Model. *Health Science Reports*, 8(5), e70726. <https://doi.org/10.1002/hsr2.70726>
3. Rahman, M. S., & **Shiddik, M. A. B.** (2025). Explainable artificial intelligence for predicting dengue outbreaks in Bangladesh

using eco-climatic triggers. *Global Epidemiology*, 100210.

4. Rahman, M. S., & Shiddik, M. A. B. (2025). Unraveling global malaria incidence and mortality using machine learning and artificial intelligence–driven spatial analysis. *Scientific Reports*, 15(1), 28334.

5. Rahman, S., & Shiddik, A. B. (2025). Reflections on explainable artificial intelligence for predicting dengue outbreaks in Bangladesh. *Global Epidemiology*, 100230.

6. Rahman, M. S., & Shiddik, M. A. B. (2025). Data-Driven Dengue Prevention Strategies in Bangladesh using Explainable Artificial Intelligence and Causal Inference. *International Journal of Statistical Sciences*, 25(2), 101-113.

7. Rahman, M. S., & Shiddik, M. A. B. (2025). Leveraging explainable artificial intelligence and spatial analysis for communicable diseases in Asia (2000–2022) based on health, climate, and socioeconomic factors. *International Journal of Health Geographics*, 24(1), 45.

8. Rahman, S. and Shiddik, A.B., 2026. Bayesian spatiotemporal modelling of neonatal, infant and under-5 mortality (2000–2022) in 41 Asian countries: a population-level observational study. *BMJ open*, 16(2), p.e108632.

9. Shiddik, M.A.B., Toshi, F.Z., Yesmin, S. and Rahman, M.S., 2026. District-Level Dengue Early Warning Prediction System in Bangladesh Using Hybrid Explainable AI and Bayesian Deep Learning. *Tropical Medicine and Infectious Disease*, 11(3), p.73.

10. Rahman, M. S., Shiddik, M. A. B., & Chowdhury, A. H. (2026). Spatiotemporal patterns of climate-sensitive vector-borne diseases in Bangladesh: leveraging machine learning and spatial regression for intervention strategies. *BMC medicine*.

11. Shiddik, M.A.B., 2026. Global Prediction of Dengue Incidence Using an Explainable Artificial Intelligence-Driven ConvLSTM Integrating Environmental, Health, and Socio-Economic Determinants. *Health Science Reports*, 9(4), p.e72280.

12. Shiddik, M.A.B., 2026. Explainable Artificial Intelligence in Healthcare: Current Landscape, Challenges, and Future Directions. *Health Science Reports*, 9(3), p.e72172.

13. Rahman, M. S., & Shiddik, M. A. B. (2026). Stillbirth, Neonatal, and Child Mortality in Bangladesh: Progress and Persistent Public Health Challenges. *Public Health Challenges*, 5(3), e70312.

REVIEWER SERVICE

- **Global Public Health** (2 Article in 2026)
- **BMJ Global Health** (2 Article in 2026)
- **BMJ public health** (2 Article in 2026)
- **Health science reports** (1 Article in 2026, 2 Article in 2025)
- **Journal of mathematics and statistics** (1 Article in 2025)
- **PLOS Digital Health** (8 Article in 2026, 6 Article in 2025)

DIGITAL PROJECTS AND TOOLS

whatdis() — R distribution detector

Developed whatdis(), a custom R package for checking and visualizing data distribution. Code and documentation available on GitHub: <https://github.com/abubokkorshiddik/R-distribution-detector> 

Dengue analytics dashboard, Bangladesh

Built an interactive Dengue dashboard that shows patterns and insights for district level dengue cases and early warning modeling in Bangladesh. Live demo:

<https://abubokkorshiddik.github.io/dengue-dashboard> 

Systematic review dashboard

Built an interactive systematic review dashboard for searching, screening, trend visualization, filtering, and result export. Live demo: <https://abubokkorshiddik.github.io/research-explorer> 

ACADEMIC CONTENT AND OUTREACH

Created educational video tutorials on Basic Statistics, R and Python for data analysis

YouTube channel: <https://www.youtube.com/@abubokkorshiddik.abs798> 

Shared public health and data science resources to promote research skills among students and early-career researchers 

REFERENCE

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